A-A-0051960(DM)
1 May 1987

INTERIM COMMERCIAL ITEM DESCRIPTION

INTRAVENOUS INJECTION SET
(Non-Vented, With Metering Chamber)

The General Services Administration has authorized the use of this interim commercial item description which was developed by the Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA, 19101, based upon currently available technical information.

This interim commercial item description covers an intravenous injection set which is non-vented and includes a metering chamber with automatic shut off.

Salient characteristics:

The intravenous injection set is a sterile, disposable, non-vented unit suitable for accurate administration of perenteral solutions in controlled volumes from 10 ml to 120 ml or up to 130 ml at 60 drops/ml from glass bottles with airways or plastic bags.

The set consists of a spike with protective cap; inlet tubing; rubber clamp; a metering chamber with a combined air valve and bacterial filter, a rubber plug injection site, and an automatic shut-off; a drip chamber with a stainless steel drip orifice to deliver 60 ± 6 drops/ml; outlet tubing; roller clamp, and flash tube and needle adapter with protective cap. The overall length shall be 84 ± 4 inches. As an alternate to the flash tube, a Y injection site 6 inches above the needle adapter will be acceptable.

All rubber and plastic components in contact with fluid being administered to the patient shall conform to USP Class IV plastic requirements for biological safety. Rubber plugs at injection sites shall be made of natural rubber and have a 45 ± 5 Shore Durometer A hardness. The tubing shall be flexible but sufficiently rigid to avoid collapse in use. The tubing shall not show exudation of plasticizers or other liquids to the surface.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The metering chamber shall be made of clear, transparent cellulose propionate seamless tubing measuring approximately 1-1/4 inches in I.D., 7-1/2 inches long with a wall thickness of 0.022 ± .004 inch. Graduations shall be present at 1 ml intervals; numbering shall be present at 10 ml intervals. The roller clamps shall be made of suitable rigid plastic and the roller shall withstand a 3.65 kg load without being removable from the clamp body.

The piercing device or spike shall be provided with an integrally molded hand grip. The spike shall have a single channel and have a conical tip suitable for leakproof insertion into solution container seals. The spike shall be 7/8 to 1 inch in length and have a base diameter of 0.208 to 0.212 inch.

The inlet tubing shall be 5 to 7 inches long, have an I.D. of 0.120 ± .005 inch and a wall thickness of 0.022 ±.005 inch. All rubber injection sites shall show no leakage except for one clinging drop at 4 psig water pressure after being punctured by an 18 gage hypodermic needle that has been thoroughly wiped to remove lubricant. All injection sites with rubber plugs shall have the plugs secured so that they are not removable in normal use. There shall be no shoulders in the bores of injection sites to deform the points of hypodermic needles.

The drip chamber shall have an effective volume of 3 to 7 ml measured from the bottom of the chamber to the outlet of the drip orifice. The drip chamber shall be transparent and have a wall that is flexible but sufficiently rigid not to undergo collapse in use.

The outlet tubing shall have an ID of 0.106 ± .005 inch and a wall thickness of 0.022 to .005 inch.

The flash tube injection site shall be molded of natural rubber with a minimum wall thickness of 0.027 inch and be 1-1/2 to 2-1/4 inches long. It shall be provided with a shoulder for injection that is a minimum of 3/8 inch in diameter and preferably a 1/2 inch in diameter. The needle adapter shall be made of suitable plastic and shall have a male Luer taper in accordance with ANSI/HIMA MD70.1. The set shall be subjected to an internal water pressure of 20 psig for 3 minutes and demonstrate no leakage or separation of components.

Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizing chamber. Each unit package shall be clearly marked with a sterility lot (control) number. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

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Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens and were found to comply with USP requirements. The certificate shall reference the method used in determining freedom from pyrogens.

Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in the USP and are found to comply with such requirements. The certificate shall reference the method used in determining safety.

Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there be any material incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

Instruction. Detailed instructions for proper use of the sets, shall be provided with each unit package.

Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing 20 individually packaged sterile intravenous injection sets, as specified, constitutes one unit (of issue). The individual package shall be a sealed, peel-open type container able to maintain sterility of contents unless package is damaged or opened. As an alternate, a sealed pouch of high density polyethylene (4 mil minimum) is also acceptable.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). of the item.

Certificates of quality. Certificates of quality supplied by the manufacturer of the components of materials may at the discretion of the contracting officer be furnished in lieu of actual performance of such testing by the contractor , provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for test purposes shall be one intravenous injection set.

TABLE I. Sampling for Examination.

Type of defect	Inspection level	AQL (percent defective)
Major	S-4	1.0
Minor	S-4	2.5

Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105 and as indicated in Table II. The unit of product shall be one IV injection set.

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Table II. Sampling for tests.

Characteristic	Inspection level	AQL (percent defective)
USP Class IV plastic **	S-2	*
Security of roller in clamp body	S-2	*
Leakage and separation of set	S-2	*
Resealing of injection sites	S-2	*
Freedom from pyrogens	S-1	*
Safety	S-1	*
Sterility	(See below)	*

* Acceptance number shall be zero for all sample sizes.

** Each lot of product need not be tested provided evidence is furnished that samples of the formulation and identical method of processing had previously been tested, that they comply with USP requirements for Class IV plastics and that they are the same as being offered on this contract.

Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination, however, shall not be restricted to the classified possible defects listed in Table III.

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TABLE III. Classification of Defects.

Categories	Defects
Major	
101	End protectors or other components missing or dislodged.
102	End protectors not properly designed to permit flow of sterilizing gases, if used.
103	Not free of bloom or tackiness.
104	End protectors not properly designed to maintain sterility.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Set not free of holes or cuts.
108	Dimensions not as specified.
109	Packaging, packing and marking not as specified.
110	Instructions missing, incomplete or illegible.
Minor	
201	Contractor's name or registered trademark, missing, incomplete, or illegible.

A. Sterility test. The intravenous injection set shall be tested for sterility in accordance with the method specified in the USP using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it is clearly differentiated from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation medium for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30^0 to 35^0 C for one half of the specimens and soybean-casein digest at 20^0 to 25^0 C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

Medium	Paper spore strip	Inoc. Prod. or Inoc. Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

*Use a minimum of 20 specimens. Acceptance number is zero.

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B. Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry Heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

C. Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in A. Sterility test and B. Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in B. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP and paragraphs A and B.

D. Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

E. Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the USP. The inspection level shall be as stated in paragraph A above.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-01-153-4703	INTRAVENOUS INJECTION SET, Non-Vented, Disposable, 20s. With Metering Chamber, 84 Inches Overall

Ordering data (Intermediate/exterior packaging quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4115

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